S2 APPENDIX

Reporting checklist and table index

STROBE checklist, an index of every result table, and the full specification history including the four abandoned analysis phases.

Supporting information for *Street slope and reported property crime in nine United States cities: terrain measurement, target exposure, and tests of candidate mechanisms.*

Companion to `PAPER.md`. Every table below is a CSV in `outputs/`, produced by the scripts

named against it. Nothing here is hand-entered.

S1. STROBE checklist (observational, cross-sectional)

#	Item	Where
1a	Study design in title/abstract	Title, Abstract
1b	Balanced summary	Abstract
2	Scientific background and rationale	§1 — three mechanisms named but untested by Haberman & Kelsay (2021)
3	Objectives, prespecified hypotheses	§1; `PREREGISTRATION.md` §2 (H1–H5)
4	Study design	§3
5	Setting, locations, periods	§3; nine US cities, incidents 2018 onward (Baltimore 2022+, Charlotte 2019-11+)
6a	Eligibility criteria	§3 "Inclusion"; `PREREGISTRATION.md` §4
7	Variables: outcomes, exposures, confounders, mediators	§3; `src/crime_classes.py`, `src/terrain.py`, `src/network.py`
8	Data sources and measurement	§3, §7; `SUBMISSION.md` §4
9	Bias	§4.6–§4.10; §6
10	Study size	Determined by data availability, not chosen — `PREREGISTRATION.md` §8
11	Quantitative variable handling	§3; slope modelled in raw degrees, TPI standardised within city

#	Item	Where
12a	Statistical methods and confounding control	§3 — Poisson PML, absorbed block-group FE, clustered SEs
12b	Subgroup/interaction	§4.3 by crime class
12c	Missing data	§4.7 coverage; exposure fallback flagged, never >0.2% of cells
12d	Sampling strategy	Full population of geocoded incidents, capped at 600k most recent per city
12e	Sensitivity analyses	§4.6 denominator, §4.7 target exposure, §4.8 spatial, §4.9 DEM resolution, §4.10 classifier
13	Participants / units	§4.1 tables; n per city in `outputs/slope_per_degree_full.csv`
14	Descriptive data	§4.1
15	Outcome data	§4.1, §4.3
16	Main results with precision	§4.1, §4.3, §4.5, §4.7 — all with 95% CIs
17	Other analyses	§4.2, §4.4
18	Key results	§5
19	Limitations	§6; `PREREGISTRATION.md` addendum D1–D11
20	Interpretation	§5
21	Generalisability	§4.1 — heterogeneity explicit, transportability disclaimed
22	Funding	None; §7
—	Ethics and responsible use	Manuscript, *Ethics and responsible use*
—	Software and AI tooling disclosure	Manuscript, *Acknowledgments*; classifier architecture in *Materials and methods*; full detail in `AI_DISCLOSURE.md` in the code repository

S2. Result tables

Headline

Table	Contents	Produced by
`slope_per_degree_full.csv`	Per-degree slope effect, all 9 cities, with floor flag	`analyze.py` + panel script
`robustness_drop_mass3.csv`	Headline excluding the contaminated loot rung	§4.10

Table	Contents	Produced by
`out_of_sample.csv`	Pittsburgh, Baltimore, Charlotte — cities that post-date variable selection	—

Mechanism tests

Table	Contents
`h1_slope_floor.csv`	Primary test: theft vs no-loot, paired cluster bootstrap, four cities
`h1_slope.csv`, `h1_noloot.csv`	Same test on wider panels and on relative height
`h2_lootmass.csv`, `loot_ladder_slope.csv`	Loot-mass ladder
`h3_mediation_multicity.csv`	Network mediation, four cities
`h4_mvt.csv`	Motor vehicle theft ranking
`h5_placebo.csv`	Sub-floor placebo cities
`asymmetry_by_class.csv`, `asymmetry_trend.csv`	Directional round-trip cost model (§4.4)

Synthesis and review-response analyses

Table	Contents
`meta_pools.csv`	Fixed-effect vs random-effects pools, τ^2, I^2 with Q-profile interval, prediction interval
`meta_regression.csv`, `meta_regression_fit.csv`	Effect on within-city slope SD — replaces the 3° threshold
`meta_leave_one_out.csv`	Pool refit without each city in turn
`ppml_diagnostics.csv`	Convergence, iterations, separated observations, singleton and all-zero block groups, sample reduction
`h1_vandalism_only.csv`	No-loot contrast with arson removed; arson estimated alone where n permits
`h1_time_matched.csv`	No-loot contrast reweighted to theft's hour × day-of-week distribution
`h1_equivalence.csv`	TOST against a margin of half the headline effect
`noloot_composition.csv`	Vandalism / arson split of the control group, by city
`temporal_split.csv`	Pre- and post-March-2020 estimates
`classifier_uncertainty.csv`, `classifier_uncertainty_pool.csv`	Label error propagated to coefficients by 200 relabelling draws
`target_multicity.csv`, `target_multicity_pool.csv`, `target_multicity_failed.csv`	Footprint target denominator; converged in 4 of 9 cities, failures recorded separately
`spatial_block_bootstrap.csv`	1 km spatial block bootstrap vs block-group clustering
`estimator_validation.csv`, `estimator_validation_failed.csv`	The same 13 models refit with `statsmodels` GLM + explicit FE dummies. Coefficients agree to 8.3e-07; SE ratios explained to 4e-05 by the finite-sample correction. Five fits that would not converge under dummies recorded separately

Robustness and measurement

Table	Contents
`spatial_diagnostics.csv`	Moran's I: all links, cross-block-group, post-filter — 14 models
`spatial_models.csv`	7 specifications × 16 models: cluster, Conley at 3 bandwidths, ESF, SEM, naive
`target_exposure_denominator.csv`	Direct test: do steep streets hold fewer targets? (No.)
`target_exposure_tests.csv`	96 rows: before/after × 2 control sets × 3 sample variants
`target_exposure_coverage.csv`	Parking-census and address coverage audit
`exposure_diagnostics.csv`	Building-footprint apportionment, 6 cities
`dem_resolution_sensitivity.csv`	1 m / 10 m / 30 m from a common lidar source
`classifier_validation.csv`	511 hand-coded offense strings
`classifier_metrics.md`	Confusion matrices, per-class precision/recall, failure patterns
`classifier_fix_impact.csv`	Class counts before and after the classifier repair

Superseded, retained for transparency

`city_estimates.csv`, `stage_two.csv`, `slope_vs_tpi.csv`, `results_*.csv`, `noloot_control.csv` — from the exploratory relative-height phase described in `DIRECTION.md`.

S3. Specification history

This project ran an extended exploratory phase before the analysis plan was written, and the treatment variable changed afterwards. Rather than present a curated path, the full sequence is recorded:

1. **Relative height (TPI), San Francisco, 100 m cells** — `PILOT_RESULTS.md`
2. **Eleven-city panel, area-apportioned exposure** — found heterogeneous and contaminated; the diagnostic that killed it is `outputs/14_signal_diagnostic.png`
3. **Two-stage decomposition** on the elevation–income correlation — `stage_two.csv`
4. **Directional round-trip cost model** — our own theory, predicting sign reversal with loot mass; unsupported (`asymmetry_trend.csv`)
5. **Slope** identified as the robust predictor; analysis plan deviations recorded in `PREREGISTRATION.md` D1–D11
6. **Confirmatory phase** — the results in `PAPER.md` §4
7. **Revision in response to external review** — the fixed-effect pool replaced by a random-effects synthesis with a prediction interval; the 3° threshold demoted from sample rule to sensitivity analysis and replaced by a continuous moderator model;

the no-loot contrast re-run without arson and with time matching, and evaluated by equivalence test rather than by a null; the target denominator extended to nine cities; classification error propagated to the coefficients; spatial block bootstrap added; estimator diagnostics reported; causal language removed throughout; two citation errors corrected and the missed literature added

Steps 1–4 are exploratory and every p-value in them is descriptive. Step 6 follows the registered rules, with the one post-hoc inclusion criterion (the gradient floor) disclosed and separately validated out-of-sample.

S4. Figures

All manuscript figures are generated by `src/figures_paper.py` against a single shared style module, `src/vizstyle.py`, which fixes the palette, the city-name mapping, legend placement and label de-collision in one place. They were previously written one at a time and had drifted apart: three different teals across figures, raw database slugs where city names belong (`montgomerycountymd`), legends floating inside the plot body, and one axis labelled "per +1 SD" on per-degree data.

The palette is a two-slot categorical set, deep petrol `#0d7d9e` against burnt rust `#c04a1e`. It was chosen for looks and then validated rather than trusted: worst-pair CVD ΔE 17.9 (protan), normal-vision ΔE 26.3, both hues inside the lightness band, above the chroma floor, and clear of 3:1 contrast against the surface. Several more muted candidates were rejected because desaturating a blue far enough to look editorial drops it under the chroma floor, where it starts reading as grey. Two series are never distinguished by colour alone: marker shape carries the same distinction, so the figures survive greyscale printing.

Figures are set in Arial at 8–9.5 pt and exported at 300 dpi, within PLOS's stated limits (Arial, Times or Symbol only, 8–12 pt, 300–600 dpi, 789–2250 px wide, ≤2625 px tall, RGB, ≤10 MB). `src/make_plos_figs.py` writes the submission TIFFs to `outputs/plos_figs/Fig1.tif`–`Fig5.tif` in citation order and fails loudly if any figure breaches a limit; the loot-ladder figure had to be narrowed from 7.66 in to 7.07 in to pass.

Seventeen superseded figures from the exploratory phases are retained in `outputs/archive/` rather than deleted, so the abandoned analyses in §S3 remain inspectable.

S5. Reproduction

`bash

python3 -m venv .venv && .venv/bin/pip install rasterio geopandas pyarrow osmnx libpysal esda spreg matplotlib statsmodels

.venv/bin/python src/registry.py # discover crime datasets

.venv/bin/python src/harvest.py # crime + DEM + terrain + census, per city

.venv/bin/python src/exposure.py # building-footprint denominators

.venv/bin/python src/network.py # street segments and network measures

.venv/bin/python src/run_confirmatory.py

`

No API keys. All sources public. Raw downloads are not redistributed but are fully re-derivable from `data/interim/registry.csv` and `registry_arcgis.csv`.